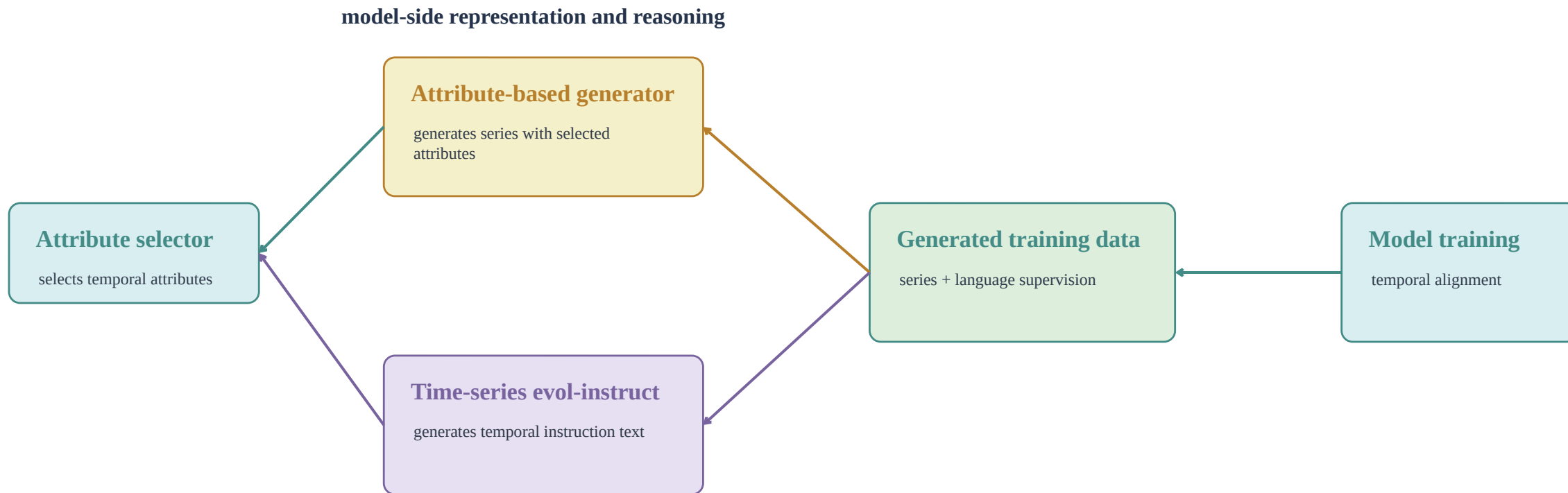


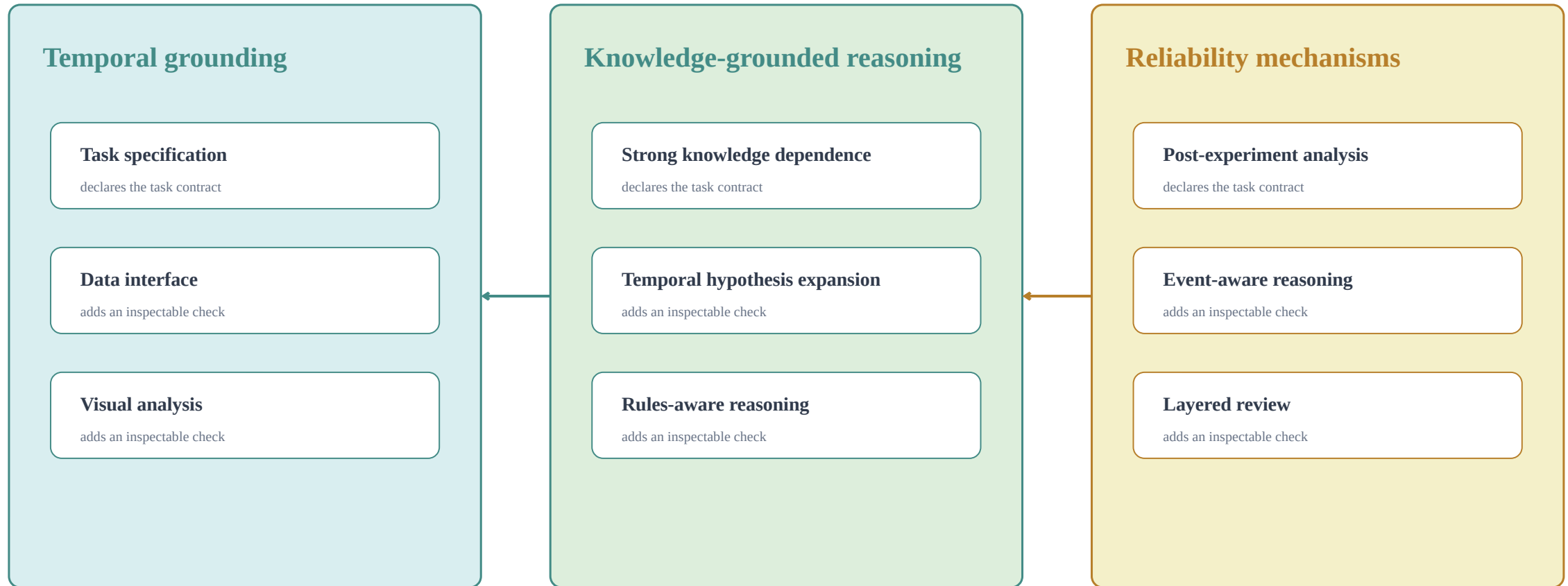
Temporal alignment and instruction data

LLM-side contribution: numerical series enter language-model training through two complementary data paths.



Three principles for a reliable time-series Agent

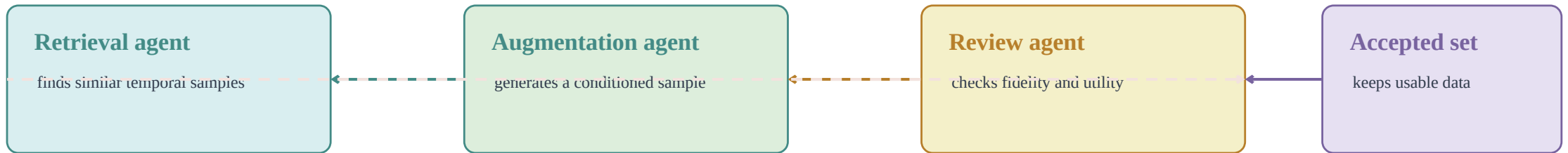
The temporal contract couples grounding, knowledge use, and reliability checks.



Retrieval, augmentation, and review

A task-specific synthesis loop couples candidate generation to fidelity and utility checks.

retrieval → **generation** → **review** → **downstream use**

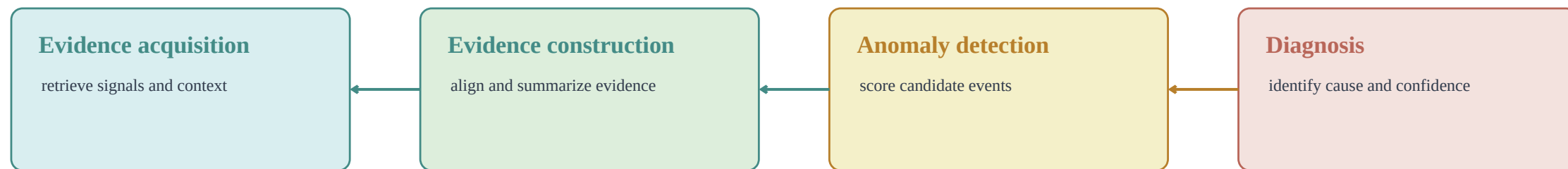


Temporal fidelity: dependence, missingness, distribution | Utility: downstream task performance

Coarse-to-fine anomaly diagnosis

Evidence acquisition and iterative refinement connect detection to a diagnosis and a final alert.

training and feedback update the detector, diagnostic tools, and review policy

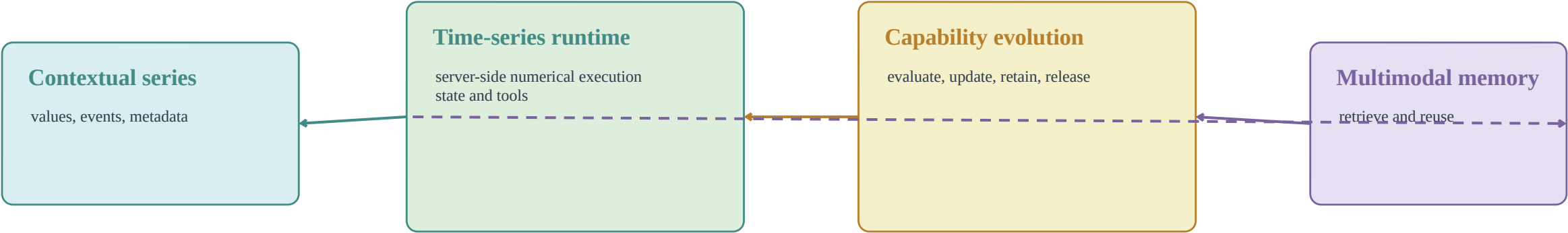


toolkit: time-series memory · numerical diagnostics · event labels · evaluator

A time-series-native runtime

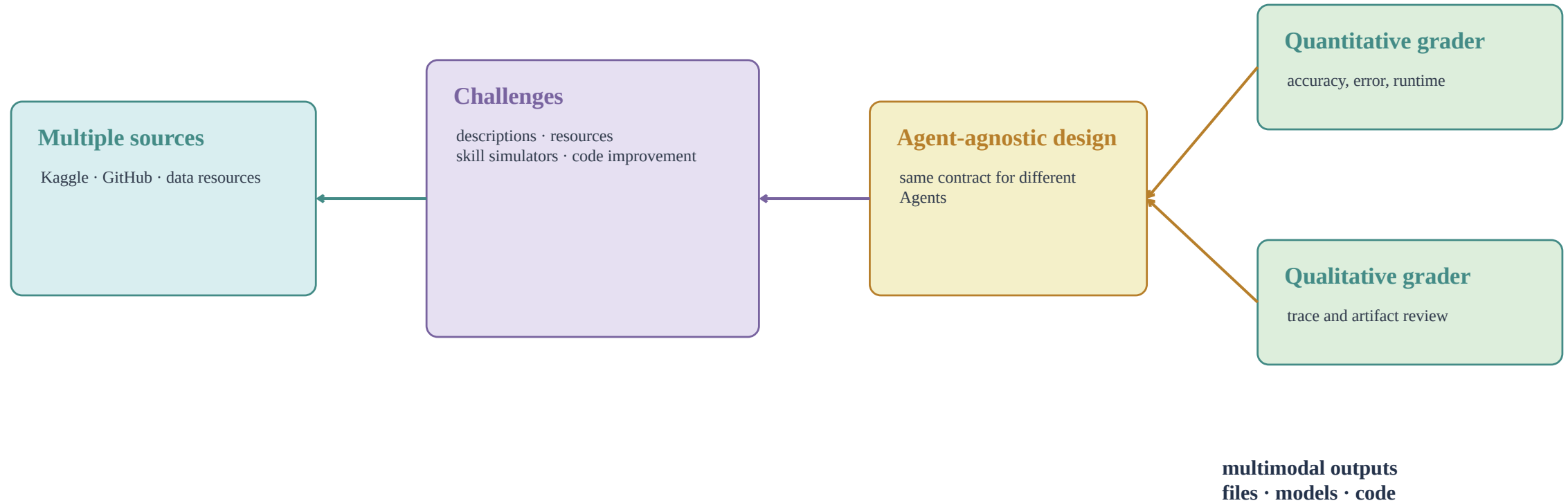
Numerical execution, capability evolution, and multimodal memory remain connected through auditable trajectories.

auditable trajectories close the loop



Benchmark infrastructure for executable agents

A benchmark combines heterogeneous sources, ML-engineering challenges, skill simulators, graders, and multimodal outputs.



Tool-using temporal reasoning trace

A General Harness loop exposes evidence, tool calls, critique, and a verified answer.

observe → **execute** → **critique** → **verify**

